

A2RL Simulator Platform

High
Fidelity
Simulation



LiDAR And GNSS/IMU Data

Actuation Commands



Steer



Brake



Throttle



Gear

CAN Signal

Localization

$$s, n, r, v_x, v_y, \beta, \zeta, \psi, \theta(s)$$

Opponent State Perception

$$\Delta s, \Delta n, v_o$$

Planning-Control
(SOC/RC-NMPC)

Game-Guided Tactical
Corridor Policy

Add Gaussian Noise

Onboard
Computer



Neosys RGS-8805GC
CPU: 64-core/ 128-thread
GPU: NVIDIA RTX A6000